THE DATA ENGINEER LAB

DATABRICKS FROM ZERO TO HERO

A 30-Day Interview & Career Readiness Series

DAY 4

Databricks Runtime (DBR) — The Engine Behind Every Cluster

SECTION 01

What Is Databricks Runtime

Databricks Runtime, commonly shortened to DBR, is the software stack that actually runs on every cluster you create. It is not just Apache Spark; it is Spark plus a set of Databricks-built enhancements bundled together as a single, versioned package.

CORE CONCEPT

Databricks Runtime is the optimized software stack that runs on Databricks clusters. It includes Apache Spark plus performance enhancements, security features, and preinstalled libraries, and different runtime versions are tuned for different workloads such as SQL, data engineering, machine learning, or streaming.

What's Actually Bundled Inside

- Apache Spark, the core distributed processing engine.
- Delta Lake, the storage layer that adds ACID transactions and reliability on top of your data files.
- Performance optimizations for I/O and query execution, tuned specifically for cloud environments.
- Security and compliance features, including encryption and fine-grained access control.
- Preinstalled and pre-configured libraries, so teams spend less time on environment setup.

SECTION 02

Types of Databricks Runtime

Not every workload needs the same software stack. Databricks offers several runtime variants, each tuned for a specific kind of job.

Runtime Type	Best Suited For
Standard Runtime	General-purpose data engineering and ETL workloads.
Databricks Runtime ML	Machine learning workloads, with libraries like TensorFlow and XGBoost preinstalled and pre-configured.
Photon-Enabled Runtime	Workloads needing a high-performance, native vectorized query engine for faster SQL and DataFrame operations.

GOOD TO KNOW

Photon is a native execution engine, not a separate product you install. When enabled on a supported runtime, it accelerates SQL and DataFrame workloads without requiring any code changes.

SECTION 03

Runtime Versioning and LTS

Databricks releases new runtime versions on a regular basis, and understanding the versioning scheme matters directly for production stability.

- Major versions: represented by the number before the decimal point, released when there are significant, sometimes non-backwards-compatible changes.
- Long Term Support (LTS) versions: marked with an LTS qualifier, offering an extended, predictable support window compared to standard releases.
- End-of-Support (EoS) versions: older runtime releases that are retired and no longer receive updates or security patches.

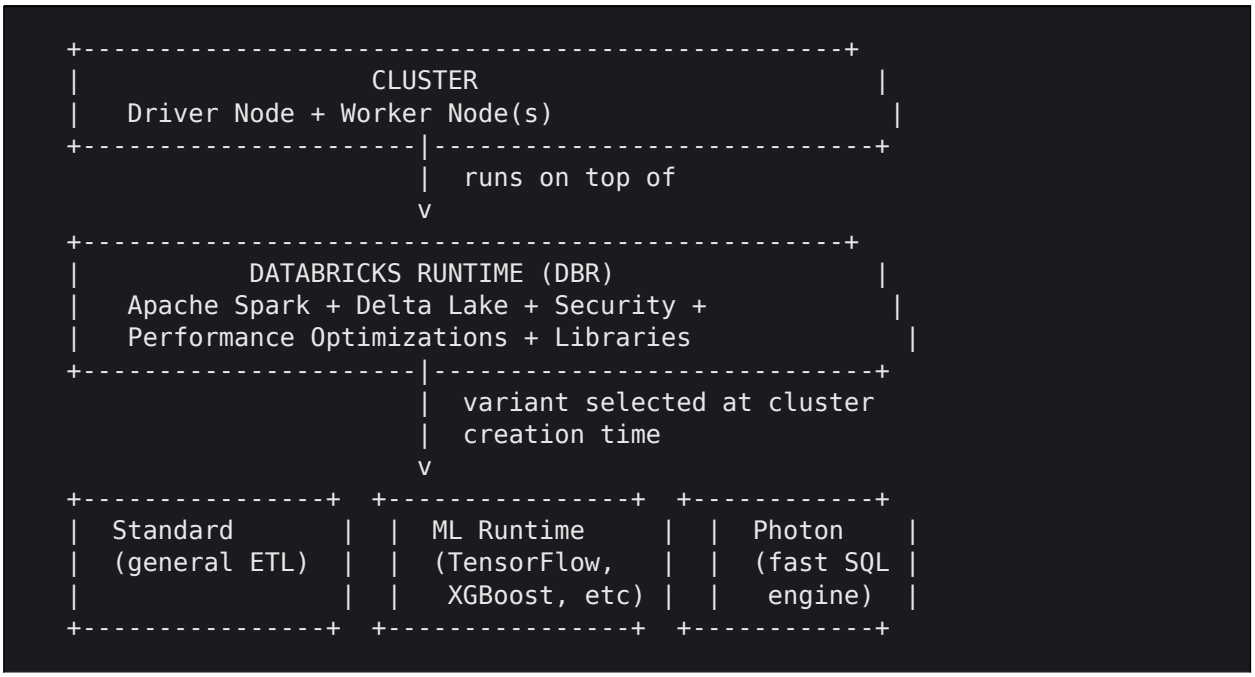
PRODUCTION BEST PRACTICE

For production clusters, choose an LTS runtime version rather than the newest available release. LTS versions get a longer, more predictable support and patch window, which matters far more for stability than being on the absolute latest version. Always check the official Databricks release notes for the current LTS recommendation, since these update over time.

SECTION 04

Where Runtime Fits in the Stack

A simplified view of how Runtime relates to the cluster and the workload running on it:



REMEMBER THIS FOR INTERVIEWS

A cluster is the compute; the Runtime is the software layer that determines what version of Spark, what libraries, and what optimizations that compute actually has available.

SECTION 05

Databricks Runtime vs Plain Apache Spark

Aspect	Open-Source Apache Spark	Databricks Runtime
Base Engine	Apache Spark	Apache Spark, Databricks-optimized
Storage Layer	Not included by default	Delta Lake included and integrated
Security Features	Configured manually	Built-in encryption, fine-grained access control
Library Setup	Manual installation required	Preinstalled, workload-specific libraries
Performance Engine	Standard Spark execution	Additional optimizations, including optional Photon
Release Cadence	Community release cycle	Faster, managed release cycle with LTS options

SECTION 06

Glossary — Day 4 Terms

Term	Definition
Databricks Runtime (DBR)	The versioned software stack, built on Apache Spark, that runs on Databricks clusters.
LTS (Long Term Support)	A runtime release track with an extended, predictable support and patch window, recommended for production.
Photon	A native, vectorized execution engine that accelerates SQL and DataFrame workloads without code changes.
Databricks Runtime ML	A runtime variant preinstalled with machine learning libraries such as TensorFlow and XGBoost.
End-of-Support (EoS)	A runtime version that is retired and no longer receives updates or security patches.

SECTION 07

Interview Questions & Answers

Q1. Is Databricks Runtime the same as Apache Spark?

A. No. Databricks Runtime includes Apache Spark but adds Delta Lake, performance optimizations, security features, and preinstalled libraries, packaged as a single versioned stack.

Q2. What is the difference between Standard Runtime, ML Runtime, and Photon?

A. Standard Runtime is for general-purpose data engineering. ML Runtime comes preinstalled with machine learning libraries. Photon is a native execution engine that can be enabled on supported runtimes to accelerate SQL and DataFrame workloads.

Q3. Why should production clusters use an LTS runtime version?

A. LTS versions offer a longer, more predictable support and patch window, which reduces the risk of unexpected breaking changes and ensures continued security patches, making them more stable for production workloads than the newest non-LTS release.

Q4. Does upgrading the Runtime version require code changes?

A. Not always, but major version upgrades can include non-backwards-compatible changes, so teams should review the official release notes and test before upgrading production clusters.

Q5. What happens if a cluster runs on an End-of-Support runtime version?

A. It stops receiving updates and security patches, which increases operational and security risk, so Databricks recommends migrating to a supported version before reaching End-of-Support.

Q6. Do you need to install Delta Lake separately on Databricks?

A. No. Delta Lake is bundled directly into Databricks Runtime, unlike open-source Apache Spark where it would need to be added manually.

STAY ON TRACK FOR ALL 30 DAYS

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